Stage	Frozen from Morocco	Re-estimated from Tetouan's unlabelled series	Changed by a single-city system
1 Per-step states and items	relative thresholds: dead ≤ 5 % of $\hat{x}$; hold and straight line ≥ 6 points	expected value $\hat{x}$ (median of the previous 14 days)	—
2 19 labelling rules	every rule and threshold	—	the two multi-city rules cannot fire
3 Label model	parameters π, θ fitted on Morocco; one E-step, no refit	—	—
4 Predictor (TCN-GAT)	architecture and training settings	weights, on rule-clean steps (two folds)	graph: the 3 same-city edges only
5 Residual hierarchy and scores	MODWT LA(8), J = 4; robust z-score, 28-day window	score scales	city part = region part
6 Event formation	w = 24, q = 99.9 %, merge gap 12, runs ≥ 3 steps	threshold γ_q at the frozen percentile	—